

Modeling the 2009 Influenza A (H1N1) Outbreak at the U.S. Air Force Academy

Introduction:

Influenza A H1N1 is a contagious respiratory illness that has not been identified as a cause of infections in people before 2009 (World Health Organization [WHO], n.d.). The H1N1's symptoms included fever, cough, sore throat, body aches, and respiratory distress, particularly affecting vulnerable groups such as children, pregnant women, and those with chronic health conditions. In 2009, the H1N1 influenza virus, also known as "swine flu," caused a global pandemic with significant health, social, and economic impacts. The first outbreak in North America was documented in April 2009, and by June 2009, a total of 74 countries and territories had reported confirmed infections (WHO, n.d.). By July 2009, more than 40,000 H1N1 cases had been confirmed, and 263 deaths in the U.S. were attributed to the H1N1 virus (Witkop et al., 2010). Given the seriousness of the disease, on July 22, clinical trials testing the 2009 H1N1 flu vaccine began and on September 15, the Food and Drug Administration approved four H1N1 2009 influenza vaccines (Centers for Disease Control and Prevention [CDC], n.d.). On December 15, the first 100 million doses of the vaccine were available for ordering. Thus, studying Influenza A provides a concrete example of how mathematical models can be used to understand transmission dynamics and inform public health responses during large-scale outbreaks.

A specific example of Influenza with a significant impact is the 2009 outbreak of Influenza A (H1N1) in the U.S. Air Force Academy in Colorado with a population of 1376 Basic Cadet Trainees. A total of 167 incident cases were reported, with 134 confirmed and 33 suspected cases of H1N1. The outbreak's onset spanned from June 26 to July 24, 2009, with peak case counts occurring on July 6, when 37 new cases were reported (Witkop et al., 2010). These figures highlight the rapid spread of the virus within the confined environment of the academy, making it a great setting for studying the dynamics of infectious disease outbreaks. In this paper, we will explore the outbreak at the U.S. Air Force Academy in detail.

The Model

To further understand the spread of Influenza A (H1N1) we can use compartmental models, such as SIR (Susceptible - Infectious - Removed) and SIJR (Susceptible - Infectious - Isolation - Removed) models. In this section, we will build two such models to simulate the Influenza A (H1N1) outbreak at the U.S. Air Force Academy in Colorado and analyze its transmission patterns over time. The first model we choose to consider is the SIR model, as it provides a simple framework for understanding the spread of the infection over time. Moreover, the SIR model helps us to estimate key parameters and allows for direct analysis of

how the disease spreads within a closed population.

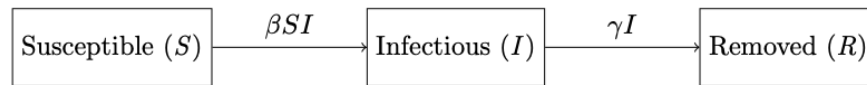


Figure 1: Compartmental diagram for SIR Model of the Influenza A (H1N1) Outbreak

In the SIR model, the transmission of Influenza A is represented through three key compartments, each associated with specific variables and parameters. The variable S represents the number of Susceptible individuals who are able to contract the disease. The variable I represents the number of Infectious individuals. Lastly, variable R represents the number of Removed individuals who have recovered from the infection.

The differential equations that help us to construct the SIR model are as follows:

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I.\end{aligned}$$

In the SIR model, parameter β represents the transmission rate of the disease, which shows how likely it is for a susceptible individual to catch the disease when they come into contact with an infectious person. This parameter helps us to estimate how quickly the disease spreads through the population. Parameter γ represents the recovery rate, indicating how quickly infectious individuals move to the removed compartment. It helps us understand the duration of the infectious period for individuals.

The second model we choose to consider is an SIJR model, an extension of the SIR model. The SIJR model incorporates incomplete isolation, providing a more detailed representation of disease spread by accounting for the part of the individuals that are infectious but isolated from the rest of the population. The SIJR model is useful for modeling the 2009 Influenza A (H1N1) outbreak at the U.S. Air Force Academy, as it accounts for a key aspect of disease transmission that was relevant during the outbreak. Specifically, the Isolation compartment of the model helps simulate the impact of public health measures, such as isolating infected individuals. By incorporating such a factor, the SIJR model provides a more thorough understanding of how Influenza spreads among the cadets.

It is important to note that we do not include an Incubation compartment in either of our two models. The incubation period for Influenza A is typically short and often does not significantly affect the overall dynamics of transmission in the context of our model. By omitting the incubation compartment, we seek to simplify the models and focus on the stages that have a more direct impact on disease spread.

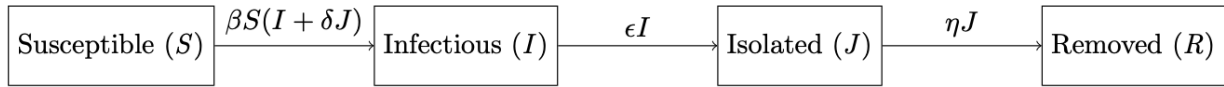


Figure 2: Compartmental diagram for SIJR Model of the Influenza A (H1N1) Outbreak

In the SIJR model, the transmission of Influenza A is represented through four key compartments. Similarly to the SIR model, the compartment S represents the number of Susceptible individuals who are able to contract the disease. The compartment I represents the number of Infectious individuals. The compartment R represents the number of Removed individuals who have recovered from the infection. Lastly, the SIJR model introduces a fourth compartment J which represents Isolated individuals. This new compartment represents individuals who, although infectious, have been isolated from the rest of the population due to public health measures, reducing their potential to transmit the disease to others.

The differential equations that help us to construct the SIJR model are as follows:

$$\begin{aligned}\frac{dS}{dt} &= -\beta S(I + \delta J) \\ \frac{dI}{dt} &= \beta S(I + \delta J) - \epsilon I \\ \frac{dJ}{dt} &= \epsilon I - \eta J \\ \frac{dR}{dt} &= \eta J.\end{aligned}$$

Similarly to the SIR model, parameter β represents the transmission rate of the disease. The parameter δ represents the influence of isolated individuals on disease transmission, meaning the parameter δ indicates the likelihood of transmission of the disease from J to S compartments. It is usually between 0 and 1, and a value closer to one indicates that transmission from the Isolation compartment is as likely as from the Infectious compartment while a value close to 0 indicates no transmission is possible from J to S compartments. Parameters ϵ and η represent how long people typically remain in each of the compartments I and J , respectively. Specifically, ϵ is the rate at which individuals move from the I compartment to the J compartment, meaning the progression from an Infectious stage to Isolation. A higher ϵ value implies a faster transition, meaning individuals spend less time in the I compartment, reducing the period during which they can spread the disease fully. Similarly, η represents the rate at which individuals exit the Isolation compartment and move into the Removed compartment, meaning it represents the recovery rate.

Data

To determine solutions to the system of differential equations that represent the SIR model, we calculate

the values of the parameters. First, we consider the recovery rate γ . We know that $\frac{1}{\gamma}$ is the average length of time an individual spends in the Infectious compartment in the compartmental diagram, meaning it is the average length of time an individual is infectious. It is stated that Influenza can be transmitted by an infected person from one day before symptoms appear until five to seven days after symptoms appear, meaning a total of six to eight days (CDC, 2024). Therefore, an individual is infectious with measles for 6 – 8 days, hence, we suppose $\frac{1}{\gamma} = 6$ days or $\frac{1}{\gamma} = 8$ days. Thus, the parameter γ is in range 0.125 – 0.167. We know that R_0 , the basic reproduction number, which represents the average number of infections produced by one infected individual in a completely susceptible population, for Influenza is between 1 and 2, meaning R_0 is in the range 1 – 2 (Virginia Department of Health [VDH], 2022). We use a parameter-based formula for R_0 to determine β . It follows that $R_0 = \frac{\beta N}{\gamma}$ where N represents the entire population. In our case, $N = 1376$, a number of Basic Cadet Trainees (BCTs) present on campus during the outbreak (Witkop et al., 2010). Thus, $1 = \frac{\beta \cdot 1376}{0.125}$, meaning $\beta = 0.000091$. Similarly, we calculate the value of β when $R_0 = 2$: $2 = \frac{\beta \cdot 1376}{0.167}$, meaning $\beta = 0.00024$. Thus, the parameter β for the SIR model is in range 0.000091 – 0.00024.

Next, we define parameters that help us construct the SIJR model. From the SIR model, we know that γ indicates the contagious period and is in the range $\frac{1}{8} - \frac{1}{6}$. We split this time into the Infectious and Isolated compartments. The time from when a person is exposed and infected with influenza to when symptoms begin is about two days (CDC, August 13, 2024). Thus, people stay in the Infectious compartment for two days and then, as they become aware of their illness, transition to the Isolated compartment. Since people stay in the Infectious compartment for two days, we set $\epsilon = \frac{1}{2}$. This leaves four to six days for people to stay in the Isolated compartment, to add up to their total of six to eight days, and so we set $\eta = \frac{1}{4}$ or $\eta = \frac{1}{6}$, meaning parameter η is in range 0.167 – 0.25. Next, it is important to notice that the greater the value of δ is, the more likely it is for people in the Isolated compartment to transmit the disease. It is stated that all cadets meeting the disease case definition were sent to a separate dorm area to convalesce until they were 7 days from symptom onset and were symptom-free for 72 hours. Although the cadets suspected or confirmed to have Influenza A H1N1 were isolated in separate dorms, the overall environment did not achieve complete isolation. Individuals were placed in designated areas, yet separation was based on symptom severity rather than ensuring absolute non-contact (Witkop et al., 2010). This approach allowed some level of interaction and transmission risk within isolated groups, meaning $\delta < 1$. Thus, a value between 0.5 and 0.8 for the parameter δ would likely capture the

moderate but not negligible risk of disease transmission within the isolated groups, as this range reflects some but not full level of interaction. Lastly, to calculate the parameter β , we use the Next Generation Method. We determine that the formula for R_0 is $R_0 = \frac{\beta N(\delta\epsilon + \eta)}{\epsilon\eta}$. Given the total population of 1376 BCTs, we can calculate the parameter β by plugging the parameters into the formula

$$1 = \frac{\beta \times 1376(0.5 \times 0.5 + 0.167)}{0.5 \times 0.167} \text{ and } 2 = \frac{\beta \times 1376(0.5 \times 0.5 + 0.167)}{0.5 \times 0.167}, \text{ meaning } \beta \text{ is in range } 0.00014 - 0.00029.$$

Results

To optimize parameters within the SIR model we use Python to compute the Residual Sum of Squares (RSS). To assess the fit of the model, we use the `odeint` function from the `scipy.integrate` package to numerically solve the differential equations representing the Influenza spread. We initially define the system of equations, which modeled the transitions between compartments (susceptible, infected, and recovered), and then adjust key parameters such as the transmission rate β and recovery rate γ . Using actual infected population data from the U.S. Air Force Academy in Colorado, we compute the RSS for multiple values β at the same time by comparing the model's predicted infected population over time with the observed values. The Python code used for computing RSS values is depicted in Figure 3 below.

```
for multiplier in range(10,30,5):
    beta = 0.00001 * multiplier
    sol = odeint(hz, y0, t_range)
    for x in range(0,20):
        new_square = (I_data[x] - sol[10*x, 1])**2
        #print("Squared distance at time", x, "=", new_square)
        RSS = RSS + new_square
    print()
    print("RSS when beta equals", round(beta,5), ":", round(RSS,4))
    print()
```

Figure 3: Python code used for RSS calculation

This code allows us to quantify the fit of the model and identify the parameter values that minimize the RSS. To find the best-fitting parameters, we iterate through various values of β and γ , testing their impact on the model's predictions. Specifically, we test the following values for β : $\beta = 0.0001$, $\beta = 0.00015$ and $\beta = 0.0002$. At the same time, we test the following values for γ : $\gamma = 0.125$, $\gamma = 0.146$, and $\gamma = 0.167$. It became clear that $\gamma = 0.146$ produces smaller RSS numbers with any β . Thus, we decided to use $\gamma = 0.146$ for our SIR model. All tested β and γ values are within biologically feasible ranges that we have determined in the Data section. Having chosen γ , we test four selected β values by iterating through them using a Python for loop. The results of the Python code are depicted in Figure 4.

RSS when beta equals 0.0001 : 1529.4406
RSS when beta equals 0.00015 : 2037.947
RSS when beta equals 0.0002 : 8245.9092

Figure 4: RSS values for multiple β parameters in a SIR model

As depicted in Figure 4, we discover that $\beta = 0.0001$ produces the smallest RSS value of approximately 1529. Based on having the smallest RSS value and being within the possible range, we decided to use $\beta = 0.0001$ for our SIR model.

Similarly, we extended this approach to the SIJR model, which includes an additional compartment for isolated individuals. By computing the RSS for parameters that are not set but rather are represented by a range, such as β , η and δ , we are able to further refine our model, ensuring that it reflects observed data from the U.S Air Force Academy more accurately. To compute the RSS we use the same python for loop as depicted in Figure 3. We test the following values for β : $\beta = 0.0002$, $\beta = 0.00021$, $\beta = 0.00022$ and $\beta = 0.00023$.

RSS when beta equals 0.0002 : 1860.0799
RSS when beta equals 0.00021 : 1844.0186
RSS when beta equals 0.00022 : 1836.6786
RSS when beta equals 0.00023 : 1845.0116

Figure 5: RSS values for multiple β parameters in an SIJR model

At the same time, we tested the following values for η : $\eta = 0.125$, $\eta = 0.146$, and $\eta = 0.167$. Additionally, we test multiple values for δ within the range $0.5 - 0.8$. The smallest RSS outcome was received with $\delta = 0.5$ and $\eta = 0.25$. Thus, these are the values that were chosen for modeling. Similarly to the SIR model, all tested β , δ and η values are within biologically feasible ranges that we have determined in the Data section. Having considered the RSS values for every value β depicted in Figure 5, we choose $\beta = 0.00022$. Having gone through the adjustment of the parameters, we create two models that represent the same Influenza A (H1N1) outbreak at the U.S. Air Academy in Colorado. Figure 6 depicts the SIR model of the outbreak, while Figure 7 depicts the SIJR model.

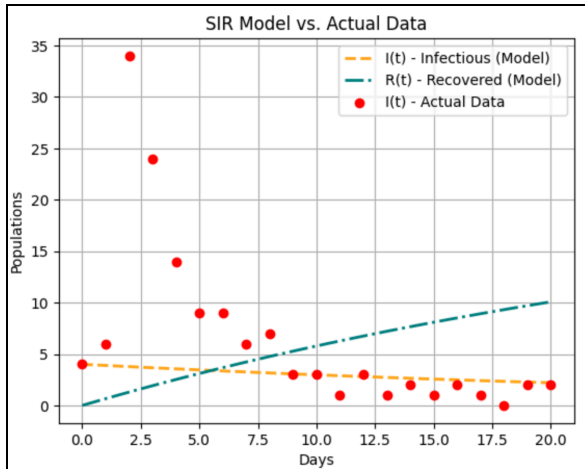


Figure 6: SIR Model of the outbreak

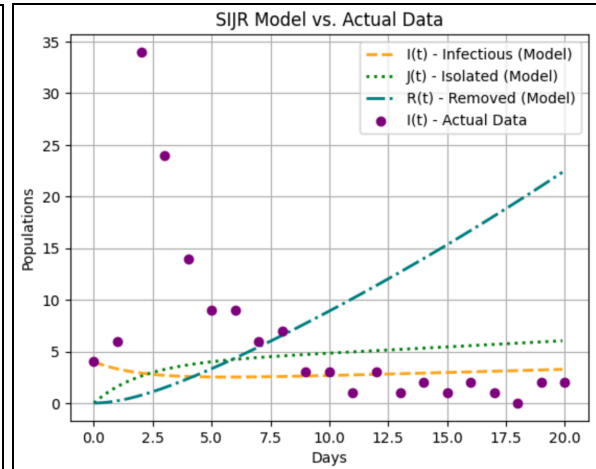


Figure 7: SIJR Model of the outbreak

It is important to note that in this analysis, we choose not to include a graph of $S(t)$ (the Susceptible population) to maintain clarity and readability in the visualization. Since $S(t)$ typically represents a large portion of the population that changes slowly compared to the other variables, such as $I(t)$, $J(t)$, and $R(t)$, its inclusion would compress the scale of $I(t)$, $J(t)$ and $R(t)$ making it difficult to mention the more subtle variations and trends. By omitting $S(t)$, the graph focuses on the dynamics of the populations with less visible changes, allowing for a clearer comparison between the model's predictions and actual data.

Discussion

In this paper, we constructed two compartmental models, the SIR and SIJR models, to simulate the dynamics of the 2009 Influenza A (H1N1) outbreak at the U.S. Air Force Academy. By comparing these models, we aim to evaluate how effectively each one captures the progression of an outbreak and analyze the differences between their assumptions and outcomes. Both models performed well in specific respects, but they differ in their inclusion and not inclusion of isolation and its effect on transmission. The SIR model, with its three compartments of Susceptible, Infected, and Recovered individuals, fit the observed data reasonably well when we optimized its parameters β and γ . However, the simplicity of the SIR model limits its ability to capture the nuances of the outbreak, especially when isolation measures are in effect. The model assumes that all individuals are either Susceptible, Infected, or Recovered, without accounting for the partial isolation implemented during the outbreak at the USAFA. As a result, it should be less accurate in representing the transmission dynamics given the present isolation.

The SIJR model includes a fourth compartment to represent Isolated individuals. This addition allows for a more realistic representation of the transmission dynamics by accounting for partial isolation, where individuals placed in the Isolation compartment could still interact with each other but at a reduced rate.

The SIJR model, therefore, should provide a closer fit to the data, as it better reflects the real-world isolation practices at the academy. Interestingly, when we compare the Residual Sum of Squares for both models, we observe that the SIR model results in a smaller RSS value, meaning that it produces outcomes closer to the real-life data. Thus, for this specific outbreak, it is not the case that the SIJR model is a better choice, as it does not fit the data as well. Additionally, in the SIR model, $I(t)$ steadily declines over time, reflecting the assumption that infected individuals recover and become immune, which naturally slows the outbreak. In contrast, the SIJR model shows an initial decrease followed by a slight increase in infections due to partially isolated individuals who still contribute to transmission. Thus, $I(t)$ fits real-life data slightly better. One possible explanation for this is the challenge of choosing the appropriate SIJR model's parameter δ which represents the likelihood of transmission from Isolated to Susceptible individuals. The value of δ is difficult to estimate, as it relies on specific details of real-world isolation practices that were not fully documented in the study. Thus, in this paper, the value of the parameter δ is a choice that was made having limited information to base it on. This limitation raises questions about how well our current modeling approach can represent transmission environments from Isolated environments. The parameter δ , while necessary for a closer fit to the data, adds uncertainty to the model, highlighting an area where additional real-world data could improve accuracy.

Additional limitations include the fact that both models were limited by the 20-day dataset of confirmed cases used in this study, which does not capture the entire outbreak period. We chose to focus only on confirmed cases to ensure the accuracy of our analysis. Confirmed cases are backed by diagnostic testing, providing a higher degree of certainty compared to suspected cases, which may introduce inaccuracies due to potential misdiagnoses or inconsistencies in reporting. Moreover, neither SIR nor SIJR models account for the sharp increase in cases on July 6, which was likely triggered by a social event on July 4, when Basic Cadet Trainees socialized with other squadrons. This case reflects a broader limitation: neither the SIR nor the SIJR model is able to represent the effects of big, high-contact events that can cause abrupt changes in transmission rates. Thus, including such events within a compartmental model would require further adjustments.

While both the SIR and SIJR models provided valuable insights into the outbreak's dynamics, the SIR model offers a closer fit to the real-life data for the 2009 Influenza A (H1N1) outbreak. Despite its simplicity, the SIR model is able to capture the main trends of the outbreak. However, the SIJR model would likely be a better choice overall, if more precise information about δ , the rate of transmission from Isolated individuals were available. Its additional complexity would allow it to account for partial isolation, which was crucial for the U.S. Air Force Academy outbreak.

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